

DIAG: Dynamics-Induced Affine Gating for Echocardiography Video Segmentation

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Abstract

Medical video segmentation must reconcile target-frame anatomy with temporal evolution. Existing methods encode both in a shared spatiotemporal state, forcing one representation to describe what anatomy is and how it changes. We instead assign them distinct modeling roles: the target frame serves as a content anchor, while temporal context parameterizes composable group transformations on that content. We propose Dynamics-Induced Affine Gating (DIAG). Single-Frame Content Anchoring preserves target-frame evidence; a Predict–Correct Latent Flow causally evolves a compact dynamics state in continuous time and corrects it with each observation; and Hierarchical Dynamic Conditioning maps this state to channel-wise affine modulation, dense spatial prompts, and region-level semantic tokens. Positive multiplicative gains form a diagonal group, realizing composable modulation of current content. Across four echocardiography datasets, DIAG outperforms the reproduced MemSAM by 0.76–1.70 Dice points on three benchmarks and reduces temporal errors by up to $12.5\times$. Controlled ablations verify each stage. Analysis reveals a near-one-dimensional cardiac-phase trajectory on CAMUS ($R^2 = .983$), while cross-dataset gain regularity and temporal-shuffling controls support temporally ordered, composable transformations.

1 Introduction

Medical video segmentation interprets each target frame using both its own observation and the way the anatomy evolves across the sequence(Quelleg et al. 2014). In echocardiography, the target frame supplies direct anatomical evidence, while the preceding sequence describes how that anatomy moves through the cardiac cycle(Sengupta et al. 2006; Causemann, Vinje, and Rognes 2022). A central modeling question is therefore how temporal evolution should be represented and brought to bear on the current frame.

A common formulation encodes historical observations and the target frame into a joint temporal state,

$$h_t = \Phi(h_{t-1}, I_t), \quad Y_t = \Psi(h_t), \quad (1)$$

where Φ updates the temporal representation and Ψ predicts the target mask. Motion-based, memory-based, recur-

rent, and continuous-state methods instantiate this formulation with different carriers and update rules(Wu et al. 2023). In all of them, historical content, current appearance, motion cues, and structural priors are aggregated into the same latent state, so temporal evolution and target-frame content share a single representation(Sun, Wang, and Xia 2026).

We adopt a different modeling role for temporal evolution. Let $x_t \in \mathcal{X}$ denote target-frame content and let temporal context parameterize an element $g_t \in G$ of a transformation group. A group action $\rho : G \times \mathcal{X} \rightarrow \mathcal{X}$ then produces the temporally conditioned representation $\rho(g_t)x_t$. Under this formulation, the target frame serves as the content anchor, latent dynamics generate transformations on that content(Wang et al. 2024), and transformations across time intervals compose through the group operation. This role separation is structural: the current observation continues to correct the dynamics state, while content and transformation retain distinct modeling roles(Pache, Battilana, and Spencer 2024).

Building on this formulation, we propose *Dynamics-Induced Affine Gating* (DIAG), which organizes video segmentation into three stages. Single-Frame Content Anchoring preserves direct evidence from the target frame. A *Predict–Correct Latent Flow* (PCLF) propagates a compact dynamics state in continuous time and corrects it with the current observation, yielding an online estimate of the transformation-generating state. *Hierarchical Dynamic Conditioning* (HDC) maps the corrected state to feature-level affine gating, dense spatial prompts, and region-level semantic conditions, modulating current-frame content at multiple levels of a pretrained segmenter. Positive multiplicative channel gains form a positive diagonal group that instantiates composable modulation, while affine biases, spatial prompts, and region-level conditions extend the conditioning range of the same dynamics state.

Across four echocardiography datasets, DIAG attains competitive segmentation accuracy together with substantially improved temporal coherence relative to frame-wise inference. Analysis further indicates that the latent dynamics organize into a low-dimensional cardiac-phase structure, and that the learned multiplicative gains exhibit consistent gain regularity and temporal-order composition across datasets, supporting temporally ordered and composable transformations. Our contributions are:

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1. We formulate video segmentation as composable transformations on target-frame content. The target frame is retained as a content anchor while temporal context parameterizes group transformations on that content, assigning current observation and temporal evolution distinct modeling roles.
2. We introduce DIAG and its Predict–Correct Latent Flow, which forms a dynamics prior through continuous prediction and corrects it with the current observation, enabling online estimation of a latent state that generates dynamic transformations.
3. We design Hierarchical Dynamic Conditioning to translate latent dynamics into feature-level affine gating, spatial prompts, and region-level semantic conditions. Positive multiplicative gains form a positive diagonal group implementing composable transformations, while the spatial and regional conditions extend the same dynamics state across representation levels of a pretrained segmenter.

2 Related Work

Medical Video Segmentation

Medical video segmentation incorporates temporal evidence through clip-level aggregation, structural propagation, or persistent state memory. Representative designs include 3D convolution, recurrent and state-space models, motion-guided propagation, and memory-based methods such as GDKVM, MemSAM, and SAM 2 (Wang et al. 2025; Deng et al. 2024; Xiong et al. 2026). Semi-supervised video object segmentation methods, including XMem, Cutie, STCN, and LiVOS, propagate object information from a ground-truth first-frame mask (Shi, Pezaros, and Elkhatib 2025; Cheng et al. 2024; Liu et al. 2025). Pretrained medical segmenters such as MedSAM and SAMed provide strong spatial priors, while subsequent variants introduce memory or temporal adaptation (Wang et al. 2023). Across these designs, temporal modeling is commonly realized through information exchange among observed frames. DIAG develops a complementary transformation-oriented formulation: the target frame remains the content carrier, while temporal context estimates how that content should be modulated.

Continuous Latent Dynamics

Neural ODEs extend discrete updates to continuous state evolution, while Latent ODEs, ODE-RNNs, and Neural CDEs combine continuous propagation with observation-time corrections (Irie, Faccio, and Schmidhuber 2022; Morrill et al. 2022). These models provide a natural representation for irregularly sampled sequences and have been explored in longitudinal imaging and physiological modeling. DIAG adopts the predict–correct principle for online video segmentation: a history-dependent flow predicts the next dynamics state, and current-frame evidence corrects that prediction before it conditions the segmenter. This separates temporal propagation from observation assimilation while retaining end-to-end learning.

Composable Conditioning

Group-based models describe structured changes through transformations that can be composed across steps, whereas FiLM, conditional normalization, and hypernetworks map external states to feature modulation (Liu, Delnevo, and Mirri 2023; Shin et al. 2022). Most equivariant constructions prescribe a geometric transformation family, while conditional modulation emphasizes adaptive feature control. DIAG connects these two directions by generating positive multiplicative gains that form a composable diagonal group. HDC further reads the same dynamics state into spatial prompts and region-level semantic conditions, allowing one temporal state to condition multiple interfaces of a pretrained segmenter.

3 Method

DIAG Framework

Given an echocardiographic video $\{I_t\}_{t=1}^T$, our goal is to predict a mask sequence $\{\hat{Y}_t\}_{t=1}^T$. The target frame supplies current content, while temporal context describes how the object evolves. DIAG assumes that temporal context can parameterize a family of composable transformations on target-frame content.

Let $x_t \in \mathcal{X}$, $z_t \in \mathcal{Z}$, and $g_t \in G$ denote target-frame content, the transformation-generating state, and the resulting dynamic transformation, respectively:

$$\begin{aligned} x_t &= \mathcal{E}_x(I_t), \\ z_t &= \mathcal{P}_{\Delta t}(z_{t-1}, \mathcal{O}(x_t)), \\ g_t &= \Gamma(z_t), \\ \hat{Y}_t &= \mathcal{D}(\rho(g_t)x_t). \end{aligned} \quad (2)$$

Here, x_t anchors current content, z_t summarizes historical evolution and is corrected by $\mathcal{O}(x_t)$, and g_t translates the resulting dynamics into a content transformation.

The map Γ generates transformations, while the representation ρ preserves their temporal composition. For $s < t < u$,

$$\rho(g_{u \leftarrow s})x = \rho(g_{u \leftarrow t} \circ g_{t \leftarrow s})x = \rho(g_{u \leftarrow t})\rho(g_{t \leftarrow s})x. \quad (3)$$

DIAG instantiates this representation using positive multiplicative gains. Let $\rho(g_t) = D_{\gamma_t} = \text{Diag}(\text{vec } \gamma_t)$ with $\gamma_t > 0$. Then

$$\begin{aligned} G_{\text{diag}} &= \{D_a \mid a \in \mathbb{R}_+^n\}, \\ D_a D_b &= D_{a \odot b}, \quad D_a^{-1} = D_{a^{-1}}, \quad D_1 = I, \\ R_{t \leftarrow s} &= D_{\gamma_t} D_{\gamma_s}^{-1} = D_{\gamma_t \oslash \gamma_s}, \\ R_{u \leftarrow s} &= R_{u \leftarrow t} R_{t \leftarrow s}. \end{aligned} \quad (4)$$

These relations provide closure, identity, inversion, and interval-wise composition for the dynamic transformation.

Figure 1 summarizes the neural instantiation. Single-Frame Content Anchoring constructs x_t , PCLF estimates z_t , and HDC generates feature, spatial, and semantic conditions.

Single-Frame Content Anchoring

For target frame I_t , a pretrained image encoder extracts a fine-scale feature and a downsampled coarse-scale feature:

$$F_t^f = \mathcal{E}(I_t), \quad F_t^c = \text{Down}(F_t^f), \quad x_t = (F_t^f, F_t^c). \quad (5)$$

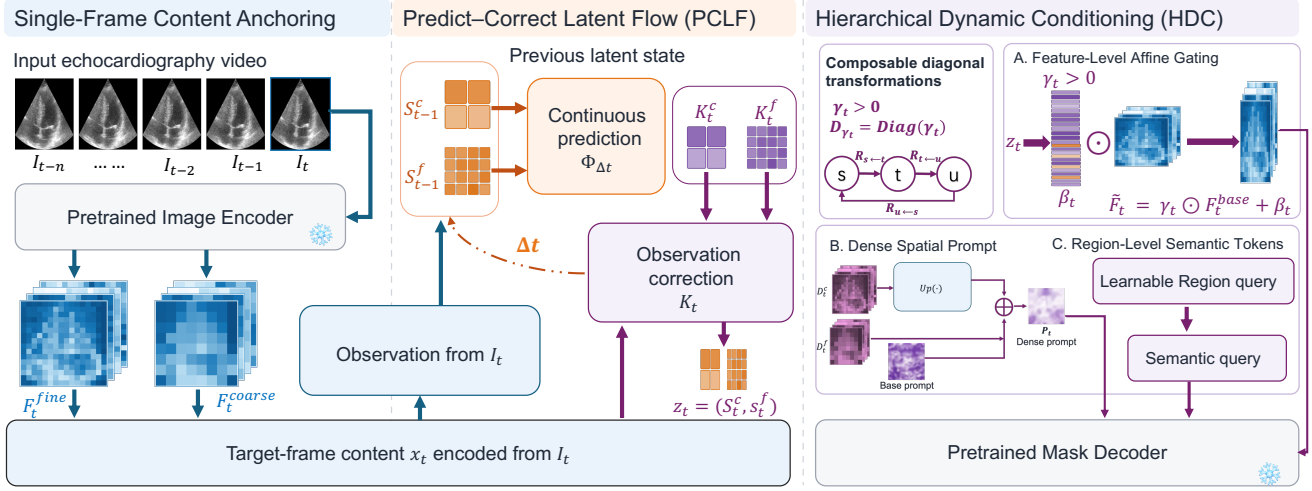


Figure 1: Overview of DIAG. Single-Frame Content Anchoring preserves a direct target-frame content path. PCLF continuously predicts dual-scale latent states and corrects them with the current observation. HDC reads the corrected dynamics into affine feature gating, dense spatial prompts, and region-level semantic tokens, which condition a pretrained mask decoder. Positive multiplicative gains instantiate composable diagonal transformations.

The fine-scale feature follows the content path toward feature gating and the mask decoder. Both scales are also supplied to the observation encoders of PCLF. Thus, the same target frame provides direct segmentation content and an observation used to correct the dynamics state.

Predict-Correct Latent Flow

PCLF takes the current dual-scale features and the preceding state, performs observation encoding, continuous prediction, and state correction, and outputs the current latent dynamics z_t .

Dual-Scale States PCLF maintains coarse and fine latent states S_t^c and S_t^f . At scale $q \in \{c, f\}$, the current feature is projected into a dynamics observation:

$$O_t^q = \mathcal{O}^q(F_t^q). \quad (6)$$

The sequence is initialized with $S_1^q = O_1^q$. The coarse state summarizes large-range evolution and the fine state retains local changes; together they define $z_t = (S_t^c, S_t^f)$.

Continuous Prediction A convolutional dynamics function f_ϕ^q parameterizes a continuous-time vector field. We use one explicit Euler step to propagate the preceding state:

$$\frac{dS^q(t)}{dt} = f_\phi^q(S^q(t)), \quad \bar{S}_t^q = S_{t-1}^q + \Delta t f_\phi^q(S_{t-1}^q). \quad (7)$$

The predicted state $\bar{S}_t^q$ depends only on history and the actual time interval and represents a dynamics prior before the current observation arrives. Explicit Δt allows one vector field to process different sampling intervals; the Euler step is a first-order neural approximation of the continuous flow.

Observation Correction PCLF concatenates the predicted state and current observation to generate a spatially

adaptive correction gate:

$$K_t^q = \sigma(\mathcal{C}^q([\bar{S}_t^q, O_t^q])), \quad (8)$$

and updates the state by

$$S_t^q = (1 - K_t^q) \odot \bar{S}_t^q + K_t^q \odot O_t^q. \quad (9)$$

The gate regulates historical prediction and current evidence at every channel and position. The corrected states form z_t and are supplied to the three conditioning branches of HDC.

Hierarchical Dynamic Conditioning

HDC maps the dual-scale state to

$$\theta_t = \mathcal{H}(z_t) = (\gamma_t, \beta_t, D_t^c, D_t^f, T_t^c, T_t^f), \quad (10)$$

where (γ_t, β_t) , (D_t^c, D_t^f) , and (T_t^c, T_t^f) condition the image feature, dense-prompt path, and mask-query path, respectively.

Feature-Level Affine Gating Let F_t^{base} denote the current content feature including the auxiliary sampling residual. Two lightweight mappings of the fine state generate channel- and position-wise scales and biases:

$$\gamma_t = \sigma(\mathcal{G}_\gamma(S_t^f)), \quad \beta_t = \mathcal{G}_\beta(S_t^f), \quad (11)$$

which modulate the target-frame feature as

$$\tilde{F}_t = \gamma_t \odot F_t^{\text{base}} + \beta_t. \quad (12)$$

The positive gain γ_t corresponds to the diagonal group element defined in Eq. (4); β_t provides additive compensation.

Spatial-Prompt Conditioning Dense heads map the two latent scales to spatial prompts D_t^c and D_t^f . After resolution alignment, they are added to the base prompt P_0 of the pretrained prompt encoder:

$$P_t = P_0 + \text{Up}(D_t^c) + D_t^f. \quad (13)$$

The coarse branch provides global regional context, while the fine branch supplies local spatial detail.

Region-Level Semantic Conditioning At each scale, K learnable region queries attend to the flattened latent state:

$$A_t^q = \text{softmax}\left(\frac{Q^q K^q (S_t^q)^\top}{\sqrt{d}}\right), \quad (14)$$

$$T_t^q = \Pi^q(A_t^q \mathcal{V}^q(S_t^q)).$$

Here, A_t^q is the attention distribution over spatial locations and $T_t^q \in \mathbb{R}^{K \times d}$ contains the resulting semantic tokens. Coarse and fine tokens are gated into the pretrained mask queries Q_0 :

$$\tilde{Q}_t = Q_0 + \alpha_c T_t^c + \alpha_f T_t^f, \quad \alpha_c, \alpha_f \in (0, 1). \quad (15)$$

The three branches jointly produce $\tilde{F}_t$, P_t , and $\tilde{Q}_t$ for the mask decoder.

Prediction and Optimization

The pretrained mask decoder predicts candidate masks and their quality:

$$\{M_t^{(j)}, u_t^{(j)}\}_{j=1}^J = \mathcal{D}(\tilde{F}_t, P_t, \tilde{Q}_t). \quad (16)$$

The highest-quality candidate is selected at inference. For annotated frames Ω , we optimize cross-entropy and Dice losses together with state reconstruction, displacement smoothness, next-frame feature prediction, and mask-quality supervision:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{flow}} \mathcal{L}_{\text{flow}} + \lambda_{\text{iou}} \mathcal{L}_{\text{iou}}. \quad (17)$$

The implementation also retains a feature-sampling residual, stopped-gradient short-range mask-token recurrence, and identity-initialized boundary refinement, which support stable training and inference while the dynamic transformation is defined by the group action above (Luo et al. 2024; Li et al. 2022). The next frame serves only as auxiliary training supervision, so inference remains online and causal. Configuration and optimization details are provided in the supplementary material (Akiba et al. 2025).

4 Experiments

Experimental Setup

We evaluate on CAMUS, EchoNet-Adult, EchoNet-Pediatric, and CardiacUDA, whose test sets contain 50, 1,276, 353, and 29 patients, respectively. All splits are patient-wise, and videos are sampled to ten frames by default. Adult, Pediatric, and CardiacUDA are scored only on frames with valid annotations. We report Dice and a unified surface-based implementation of HD95; centroid drift, area roughness, and inter-frame Dice variation characterize temporal behavior. When multiple tuned runs exist, we select

a complete checkpoint using validation Dice, and all metrics in one table row come from that checkpoint. DIAG uses neither ground-truth masks nor box prompts at test time. Architecture, training, and metric details are provided in the supplementary material (Ullah et al. 2025; Tromp et al. 2022; Yang et al. 2023).

The evaluation follows four hypotheses: target-frame content anchoring (H1), evolving predict-correct states (H2), complementary hierarchical readouts (H3), and temporally ordered gain composition (H4). Main comparisons establish practical utility, followed by component removals, sequence perturbations, state interventions, and composition controls.

Comparison with Existing Methods

Table 1 compares DIAG with automatic temporal models and prompted video-segmentation references. On CAMUS, Adult, and Pediatric, DIAG ranks second in Dice, trailing the best method by only 0.08, 0.04, and 0.07 percentage points, respectively. On the more domain-shifted CardiacUDA dataset, it achieves the best automatic Dice of .8092, exceeding the second-best result by 3.05 points. Among automatic methods with complete coverage of all four datasets, DIAG obtains the highest mDice of .8911, improving over GDKVM by 1.99 points.

DIAG also maintains a strong balance between overlap and boundary accuracy. Relative to GDKVM, it reduces HD95 from 8.27 to 5.75 on Adult and from 4.80 to 4.25 on Pediatric, corresponding to reductions of 30.5% and 11.5%, while their Dice scores differ by less than .001. On CAMUS, DIAG ranks second in both Dice and HD95, whereas ConvLSTM attains the lowest HD95. These results show that strong overlap does not always translate into equally accurate boundaries.

The ranking varies across datasets. GDKVM performs best on the three primary benchmarks but drops markedly on CardiacUDA, while convolutional and recurrent models exhibit dataset-specific strengths. DIAG remains close to the best Dice on the primary benchmarks and performs most strongly under domain shift, yielding the highest complete-domain macro-average. This pattern is consistent with its design, where target-frame content anchors segmentation and temporal dynamics modulate its interpretation. Prompted methods are reported only as references because they use ground-truth initialization at test time.

Figure 2 complements the quantitative comparison with paired ED-ES frames from the same patient. DIAG preserves the principal region and contour on the selected Adult and Pediatric median cases, while GDKVM performs better on the selected CAMUS case, consistent with its slightly higher aggregate Dice. Because the cases are fixed by patient-level median Dice before visual inspection, they reflect typical rather than selectively favorable performance. The remaining DIAG errors are mainly contour-local, in agreement with its stronger HD95 results on Adult, Pediatric, and CardiacUDA.

Verifying the Three-Stage Design

Removing the entire temporal path decreases Dice by 2.63 points, establishing the value of dynamics conditioned on a single-frame anchor. Within PCLF, removing continuous propagation or dual-scale states costs 0.39 and 0.19 points;

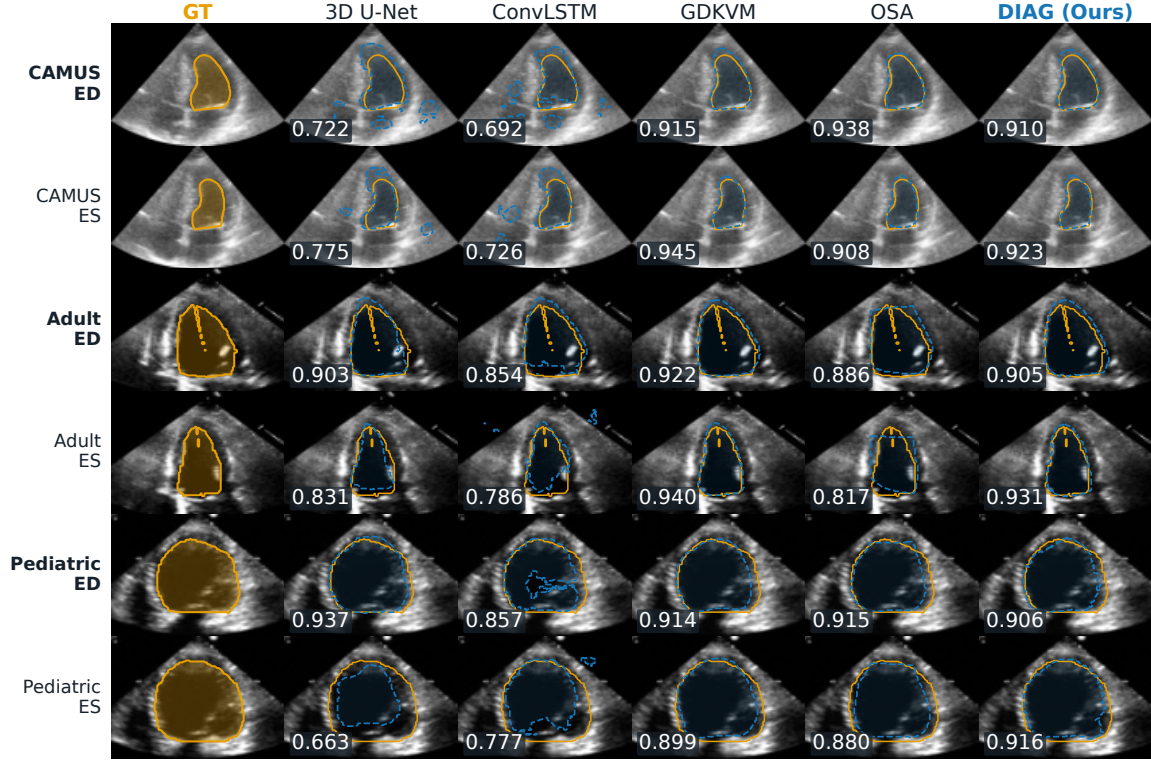


Figure 2: Automatic ED-ES segmentation on matched cases. Each case is fixed at the dataset-median patient-level DIAG Dice before visual inspection. Orange solid and blue dashed contours denote ground truth and prediction; numbers are frame-level Dice scores.

within HDC, removing spatial prompts, affine gating, and region-level conditioning costs 0.78, 0.46, and 0.17 points, respectively. The three conditioning interfaces are complementary—the spatial prompt contributes the largest localization correction while affine gating realizes a multiplicative transformation on current content—and removing all temporal conditioning produces a substantially larger loss than removing any single component, so the gain reflects the full temporal path rather than one adapter alone. Because each row is a single-component removal, these differences are interpreted as relative contributions rather than additive effects; integrator and conditioning alternatives are deferred to the supplementary material (Ingendahl et al. 2024).

The frame-only comparison establishes the value of the complete temporal path, and the targeted affine removal together with the state-intervention and composition controls in Figure 3 further identify composable modulation as its source; the ablation and mechanism analyses thus answer complementary questions—whether each component improves segmentation, and whether the resulting representation behaves as proposed. Together with the prompt-free cross-domain results, the 2.63-point frame-only gap supports H1, and the consistent degradation from all three HDC removals supports H3, with their unequal magnitudes indicating that hierarchical conditioning is complementary.

Temporal Dynamics beyond Frame-wise Accuracy

Figure 4 shows that DIAG improves patient-level Dice by 1.25 and 1.16 points on Adult and Pediatric while reducing centroid drift by 41.7% and 32.4%. On CAMUS, frame-wise inference exhibits $7.9\times$, $2.3\times$, $12.5\times$, and $3.8\times$ the Dice variation, centroid jitter, area roughness, and accumulated drift of DIAG. Temporal modeling therefore improves not only mean overlap but also trajectory stability.

The four CAMUS measures capture overlap fluctuation, spatial displacement, area change, and accumulated deviation. Their agreement, together with simultaneous Dice gains on Adult/Pediatric, indicates improved trajectories rather than temporally conservative masks.

Figure 5 evaluates changes in observation conditions. At five frames, every dataset retains at least 98.9% of its ten-frame Dice. Under three frames, 70% random observation removal, and 40% timestamp jitter, CAMUS retains 95.9%, 97.0%, and 97.5%, respectively. PCLF thus does not rely on a fixed ten-frame template and remains stable under sparse observations and altered intervals.

These tests perturb when evidence is available rather than measuring smoothness under default sampling. Because the same checkpoint is used without adaptation, the retained accuracy supports interval-aware state propagation. Together with the coherence gains, these results support the operational role of an evolving temporal state (H2); its structure

Method	Temporal Modeling	Test Cue	CAMUS		Adult		Pediatric		CardiacUDA		mDice $\uparrow$
			Dice $\uparrow$	HD95 $\downarrow$	Dice $\uparrow$	HD95 $\downarrow$	Dice $\uparrow$	HD95 $\downarrow$	Dice $\uparrow$	HD95 $\downarrow$	
3D U-Net	3D convolution	None	.9162	5.46	.8468	6.16	.8436	7.09	.7787	6.29	.8516
ConvLSTM	Recurrent state	None	.9314	3.00	.8475	6.48	.8382	7.29	.7518	10.17	.8422
GDKVM	KV memory	None	.9368	6.13	.9150	8.27	.9052	4.80	.7279	7.38	<u>.8712</u>
MemSAM	SAM memory	None	.9284	—	.8976	8.33	.8887	5.37	—	—	—
OSA	Orthogonal state	None	.9284	5.83	.7740	10.72	.8186	4.17	.5948	—	.7790
SAM2-memory	Memory attention	None	.9212	6.79	.8915	7.63	.8823	7.34	—	—	—
DIAG	Predict-correct flow	None	<u>.9360</u>	<u>5.43</u>	<u>.9146</u>	5.75	<u>.9045</u>	<u>4.25</u>	.8092	6.18	.8911
<i>Prompted references (not ranked with automatic methods)</i>											
XMem †	Object memory	GT mask	.9146	7.12	.8441	7.12	.8130	8.27	.8437	6.03	.8538
Cutie †	Object memory	GT mask	.9112	7.17	.8426	7.48	.8108	5.58	.8034	4.57	.8420
STCN †	Correspondence memory	GT mask	.8767	11.59	.8386	8.27	.8375	8.47	—	—	—
LiVOS †	Long-term memory	GT mask	.8364	10.42	.7081	8.82	.7011	9.13	.6439	11.26	.7224
SAMed-2 ‡	Prompted SAM2	GT box	.8496	7.16	.5823	10.13	.6235	12.86	.1789	9.63	.5586

Table 1: Segmentation results on four echocardiography datasets. The upper block follows a fully automatic protocol; bold and underline indicate its best and second-best results. mDice is the unweighted macro-average over four datasets and is reported only for methods with complete coverage. The lower block uses a ground-truth first-frame mask (†) or ground-truth box (‡) and is excluded from automatic ranking. “—” denotes no trustworthy result under the same protocol.

Variant	Flow	MS	Aff.	Spa.	Reg.	ED $\uparrow$	ES $\uparrow$	Mean $\uparrow$
Frame-only						.9121	.8987	.9054
w/o spatial	✓	✓	✓		✓	.9123	.9355	.9239
w/o affine	✓	✓		✓	✓	.9386	.9143	.9271
w/o flow		✓	✓	✓	✓	.9376	.9165	.9278
Single-scale	✓		✓	✓	✓	.9350	.9246	.9298
w/o region	✓	✓	✓	✓		.9407	.9178	.9300
DIAG	✓	✓	✓	✓	✓	.9427	.9218	.9317

Table 2: Controlled component ablation on CAMUS. ED and ES report endpoint Dice, while Mean averages the ten sampled frames. Each variant removes one component from the same full configuration.

and decoder use are tested next.

Latent-State Use and Composable Transformations

Figure 6 aligns PCLF correction and two spatially visualizable HDC responses on the same cases. State correction localizes where current evidence revises the dynamics prior; affine conditioning changes target-feature responses; and the spatial branch produces a dense decoder prompt. Region-level tokens are not rendered as heatmaps because they do not share the spatial representation of the other branches.

The three preselected performance quantiles prevent this mechanism view from collapsing to one successful case. More importantly, the branches exhibit different spatial roles on aligned inputs: correction revises the propagated state, affine response redistributes feature emphasis, and the prompt supplies localized decoder evidence. Their visual difference is descriptive, while their necessity is established by the controlled removals in Table 2. Together, these results support H3: a shared dynamics state is read out into feature

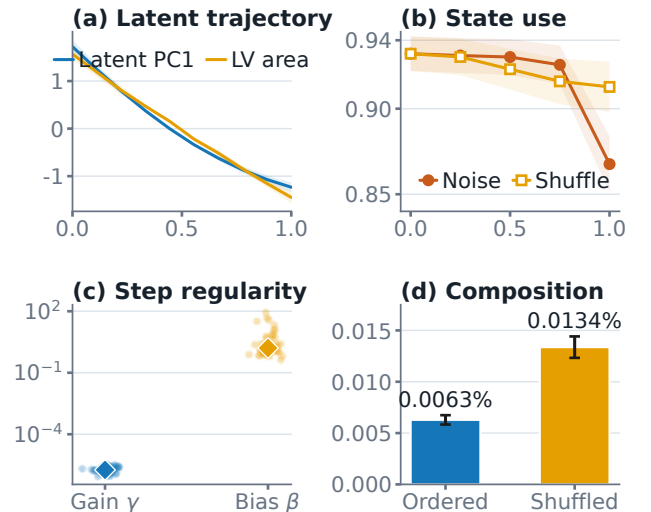


Figure 3: Dynamics and transformation diagnostics. (a) Latent/area trajectory ($n = 50$); (b) state interventions ($n = 25$); (c) adjacent-step CV of γ and β ; and (d) ordered versus shuffled composition (450 comparisons). Shading and error bars denote 95% CIs.

reweighting, localized evidence, and semantic conditioning rather than copied into equivalent adapters.

Continuous prediction reduces the mean adjacent-state change from 43.29 for direct observations to 17.97 for the dynamics prior; after observation correction it remains 18.59. Figure 3(a,b) further shows that corrected CAMUS states form a low-dimensional trajectory synchronized with left-ventricular area and linearly decode cardiac phase with $R^2 = .983$. Cross-time shuffling and noise injection re-

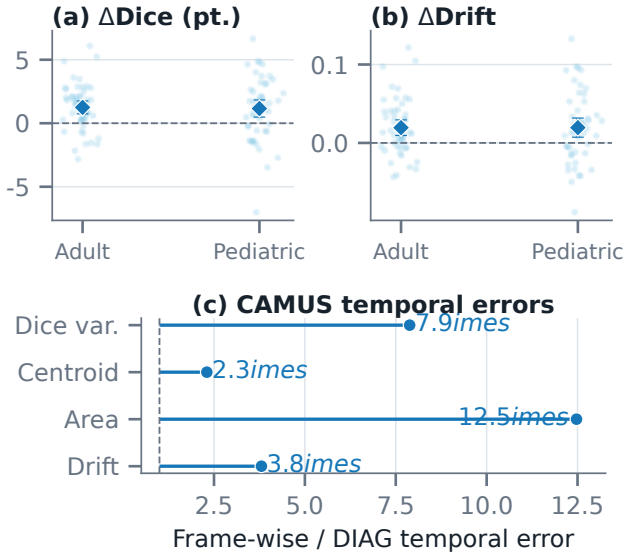


Figure 4: Temporal benefit. Patient-level Dice gains and centroid-drift reductions over frame-wise inference on Adult/Pediatric ($n = 50$ per group), and four ratios of frame-wise to DIAG temporal error on CAMUS. Light points are patients; diamonds and bars show means and 95% CIs.

duce Dice to .9127 and .8676, indicating that downstream segmentation uses the state.

These controls distinguish a functional dynamics state from a merely correlated or overly smooth representation. The intervention drop shows that segmentation depends on the state while target-frame content is fixed, whereas the small increase from 17.97 to 18.59 after correction shows that current evidence is incorporated without resetting the temporal trajectory.

Figure 3(c,d) examines the multiplicative branch. Across three datasets, the adjacent-ratio CV of γ is on the order of 10^{-6} , whereas additive bias lacks comparable step regularity. Ordered composition yields .0063 error; temporal shuffling increases it to .0134, or $2.13\times$. Together with closure and invertibility under positive diagonal multiplication, these observations support a compact, regular, and order-sensitive empirical transformation trajectory. Additional participation-ratio, extrapolation, and cross-domain diagnostics are provided in the supplementary material (Yuan et al. 2025).

Overall, phase alignment, state intervention, prediction–correction statistics, and composition tests support H2 at the state level and H4 at the generated-transformation level, matching the roles of z_t and $g_t = \Gamma(z_t)$ in Eq. (2). Taken together, these analyses show that DIAG learns a functional dynamics state whose evolution is both temporally structured and directly used by the decoder.

5 Conclusion

We introduced DIAG, which models temporal evolution as composable transformations of target-frame content. Its predict–correct latent flow estimates dynamics online, while

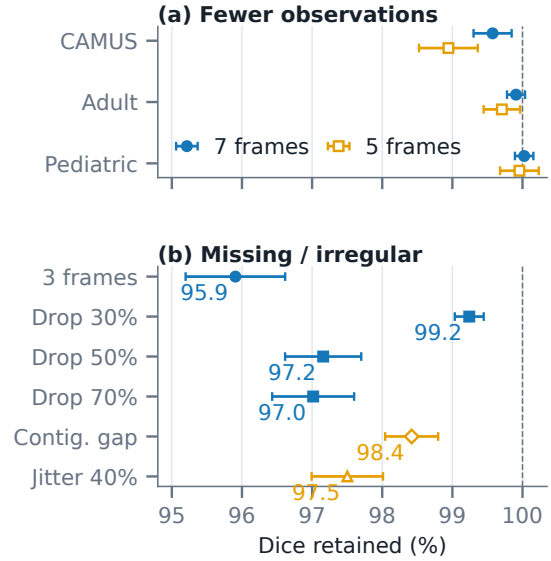


Figure 5: Sampling stress tests. Dice retention when reducing three datasets from ten to seven/five frames, and performance of one CAMUS checkpoint under sparse observations, contiguous gaps, and timestamp perturbations. Error bars are patient-level 95% CIs;

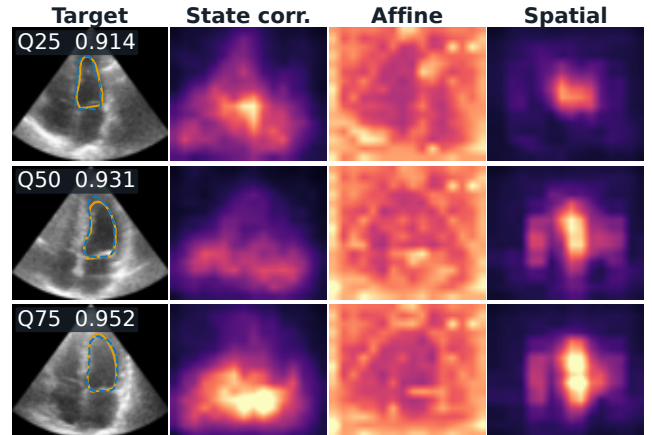


Figure 6: PCLF/HDC responses on CAMUS cases preselected at the 25th, 50th, and 75th Dice percentiles. Columns show the target frame, state correction, affine response, and spatial prompt. Orange/blue contours denote ground truth/DIAG;

hierarchical conditioning applies the resulting state across feature, spatial, and semantic levels. Experiments on four echocardiography datasets demonstrate competitive segmentation, improved temporal consistency, and robustness to sparse sampling. These results show that temporal evolution can effectively condition the current frame while retaining target-frame content as the segmentation anchor.

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